Strain

Strain is defined as the relative change in the position of **points** within a body that has undergone deformation.

Normal strain (ε_{ii}): It is the ratio of the change in length to original length of a linear element considered in the body due to the deformation.

Shear strain(γ_{ij}): It is the reduction in 90^0 angle between two mutually perpendicular lines considered in the body due to deformation.

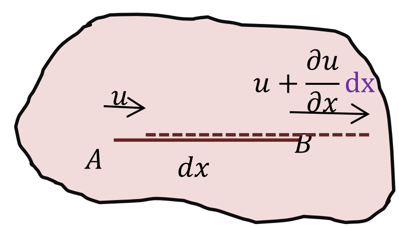
The strain-displacement relation:

A body undergoes a strain because there are relative displacements between different points in the body when it is subjected to load. If there is no relative displacement, the body undergoes rigid body motion.

Consider a line AB of length dx oriented along the x-direction. Let u be the displacement component in x-direction at point A. Then the displacement of point B which is at a distance dx is $u + \frac{\partial u}{\partial x}\,dx$.

The change in length of AB $= u + \frac{\partial u}{\partial x}\,dx - u = \frac{\partial u}{\partial x}\,dx$

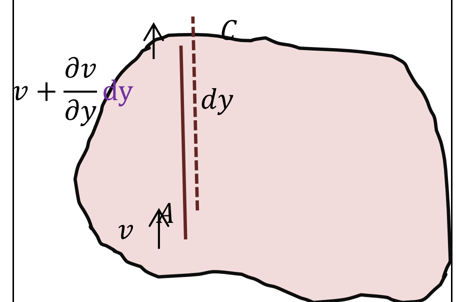
$\therefore$ the normal strain in x-direction, $\varepsilon_{xx} = \frac{\frac{\partial u}{\partial x}dx}{dx} = \frac{\partial u}{\partial x}$



Consider another line AC of length dy oriented along the y-direction. Let u be the displacement component in y-direction at point A. Then the displacement of point C which is at a distance dy is $v + \frac{\partial v}{\partial y}\,dy$.

The change in length of AC $= v + \frac{\partial v}{\partial y}\,dy - v = \frac{\partial v}{\partial y}\,dy$

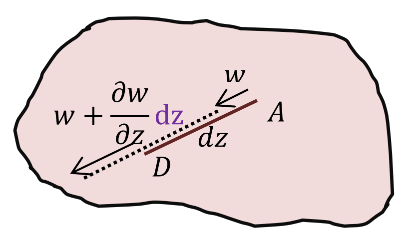
$\therefore$ the normal strain in y-direction, $\varepsilon_{yy} = \frac{\frac{\partial v}{\partial y}dy}{dy} = \frac{\partial v}{\partial y}$



Similarly, along z-direction, let w be the displacement component in z-direction at point A. Then the displacement of point D which is at a distance dz is $w + \frac{\partial w}{\partial z}\,dz$.

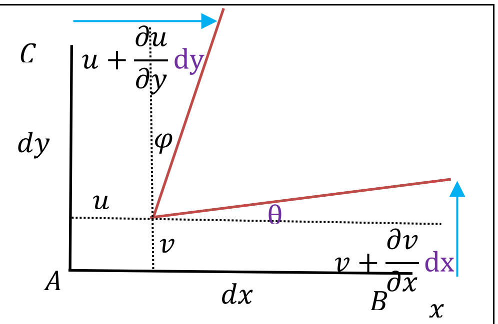
The change in length of AD $= w + \frac{\partial w}{\partial z}\,dz - w = \frac{\partial w}{\partial z}\,dz$

$\therefore$ the normal strain in z-direction, $\varepsilon_{zz} = \frac{\frac{\partial w}{\partial z}dz}{dz} = \frac{\partial w}{\partial z}$



Shear strain (γ_{xy})

Consider the perpendicular lines AB and AC together. We have considered the normal strains happening to these lines. In addition to the x- displacement, the line AC has displacement in the y-direction also. Because of this, the end B moves to a distance $v + \frac{\partial v}{\partial x}dx$.



So, $\tan\theta = \frac{v + \frac{\partial v}{\partial x}dx - v}{dx} = \frac{\partial v}{\partial x}$. For small deformation, $\tan\theta \cong \theta$

$$\theta = \frac{\partial v}{\partial x}$$

Similarly, line AC not only has y-displacement, but a x-displacement $u + \frac{\partial u}{\partial y}dy$ also.

$$\therefore \tan\varphi \cong \varphi = \frac{u + \frac{\partial u}{\partial y}dy - u}{dy} = \frac{\partial u}{\partial y}$$

The shear strain is defined as the reduction in the $90°$ angle between two perpendicular lines due to deforation.

$$\gamma_{xy} = \theta + \varphi = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

Similarly, when we consider the perpendicular lines AC and AD (in the y-z plane)

$$\gamma_{yz} = \frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}$$

And AB and AD (in the x-z plane)

$$\gamma_{zx} = \frac{\partial w}{\partial x} + \frac{\partial u}{\partial z}$$

We can notice that shear strain $\gamma_{xy} = \gamma_{yx}$, $\gamma_{yz} = \gamma_{zy}$, and $\gamma_{zx} = \gamma_{xz}$.

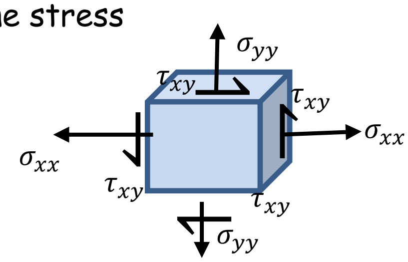
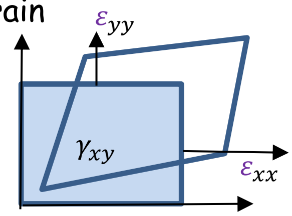
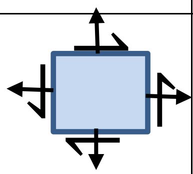
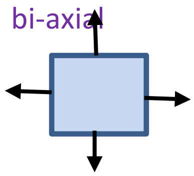
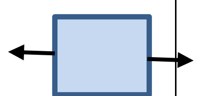
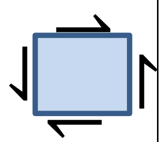
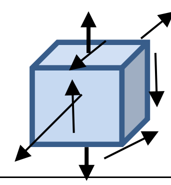
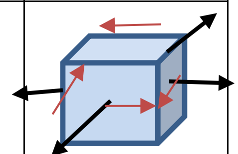
We can write the strain tensor as

$$\varepsilon_{ij} = \begin{bmatrix} \varepsilon_{xx} & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_{yy} & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_{zz} \end{bmatrix}$$

Where $\varepsilon_{xy} = \frac{\gamma_{xy}}{2}$, $\varepsilon_{yz} = \frac{\gamma_{yz}}{2}$, and $\varepsilon_{xz} = \frac{\gamma_{xz}}{2}$

Plane stress and Plane strain

In many engineering applications, the state of stress or strain at a point can be represented in one plane (i.e., no stress or strain components in the third direction). Such a system is known as plane stress or plane strain. We will see later that both plane stress and plane stain conditions cannot exist simultaneously.

<table>
<tr><td colspan="3">Eg. plane stress
</td><td colspan="3">plane strain
</td></tr>
<tr><td>$\begin{bmatrix}\sigma_{xx} & \tau_{yx} & 0\\ \tau_{xy} & \sigma_{yy} & 0\\ 0 & 0 & 0\end{bmatrix}$</td><td>$\begin{bmatrix}\sigma_{xx} & 0 & 0\\ 0 & \sigma_{yy} & 0\\ 0 & 0 & 0\end{bmatrix}$</td><td>$\begin{bmatrix}\sigma_{xx} & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$</td><td>$\begin{bmatrix}0 & \tau_{yx} & 0\\ \tau_{xy} & 0 & 0\\ 0 & 0 & 0\end{bmatrix}$</td><td>$\begin{bmatrix}0 & 0 & 0\\ 0 & \sigma_{yy} & \tau_{zy}\\ 0 & \tau_{yz} & \sigma_{zz}\end{bmatrix}$</td><td>$\begin{bmatrix}\sigma_{xx} & 0 & \tau_{zx}\\ 0 & 0 & 0\\ \tau_{xz} & 0 & \sigma_{zz}\end{bmatrix}$</td></tr>
<tr><td></td><td>bi-axial
</td><td>uni-axial
</td><td>pure shear
</td><td></td><td></td></tr>
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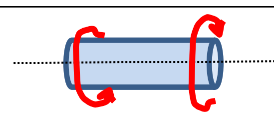
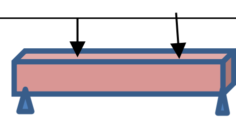
The problems in solid mechanics involve

1. Axial loading of bars		Normal stresses on cross section (Tensile or compressive)
2. Torsional loading of circular shafts		Shear stresses on cross section
3. Bending (Flexural/Transverse) loading of beams		Normal and shear stresses
4. Combined loading (Axial+bending+torsion)		